Chapter 9. The Fundamental Group

Path homotopy, covering spaces, the circle, retractions and fixed points, Borsuk–Ulam, homotopy type, spheres and surfaces

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9.1 Homotopy of paths

Maps $f, f' : X \to Y$ are **homotopic**, $f \simeq f'$, if there is a continuous $F : X \times I \to Y$ ($I = [0,1]$) with $F(x,0) = f(x)$ and $F(x,1) = f'(x)$; f is **nulhomotopic** if it is homotopic to a constant map. Paths $f, f' : I \to X$ from x_0 to x_1 are **path homotopic**, $f \simeq_p f'$, if there is $F : I \times I \to X$ with $F(s,0) = f(s)$, $F(s,1) = f'(s)$, $F(0,t) = x_0$, $F(1,t) = x_1$.

Lemma 9.1 (51.1)

$\simeq$ and $\simeq_p$ are equivalence relations.

Proof. $F(x,t) = f(x)$ gives $f \simeq f$; if F is a homotopy from f to f', then $F(x, 1-t)$ is one from f' to f. If $F : f \simeq f'$ and $F' : f' \simeq f''$, define $G(x,t) = F(x,2t)$ for $t \le \frac{1}{2}$ and $F'(x, 2t-1)$ for $t \ge \frac{1}{2}$; the two formulas agree at $t = \frac{1}{2}$, so G is continuous by the pasting lemma. For paths, all these homotopies keep the end points fixed. ∎

We write $[f]$ for the path-homotopy class of f.

Example 9.1

If $f, g : X \to \mathbb{R}^2$ are maps (or paths with the same end points), the **straight-line homotopy** $F(x,t) = (1-t)f(x) + tg(x)$ is a homotopy (path homotopy). More generally this works for maps into any convex subset of $\mathbb{R}^n$. In $\mathbb{R}^2 - \{0\}$, however, the paths $f(s) = (\cos \pi s, \sin \pi s)$ and $g(s) = (\cos \pi s, -\sin \pi s)$ are not path homotopic (Section 9.4).

If f is a path from x_0 to x_1 and g a path from x_1 to x_2, the **product** $f * g$ is $f(2s)$ on $[0, \frac{1}{2}]$ and $g(2s-1)$ on $[\frac{1}{2}, 1]$. It is well defined on classes: if $F : f \simeq_p f'$ and $G : g \simeq_p g'$, then $H(s,t) = F(2s,t)$ for $s \le \frac{1}{2}$, $G(2s-1,t)$ for $s \ge \frac{1}{2}$, is a path homotopy $f * g \simeq_p f' * g'$. So $[f] * [g] = [f * g]$ is defined. Two observations will be used repeatedly: if $k : X \to Y$ is continuous and F is a path homotopy from f to f', then $k \circ F$ is a path homotopy from $k \circ f$ to $k \circ f'$; and $k \circ (f * g) = (k \circ f) * (k \circ g)$.

Theorem 9.2 (51.2)

The operation $*$ on path-homotopy classes satisfies:
(1) (associativity) if $[f] * ([g] * [h])$ is defined, so is $([f] * [g]) * [h]$, and they are equal;
(2) (identities) if e_x is the constant path at x and f runs from x_0 to x_1, then $[e_{x_0}] * [f] = [f] = [f] * [e_{x_1}]$;
(3) (inverses) with $\bar{f}(s) = f(1-s)$, $[f] * [\bar{f}] = [e_{x_0}]$ and $[\bar{f}] * [f] = [e_{x_1}]$.

Proof. Let e_0 be the constant path in I at 0, $i : I \to I$ the identity path, and $\bar{i}$ its reverse.

(2) $e_0 * i$ and i are paths in the convex set I from 0 to 1, so they are path homotopic by a straight-line homotopy G; then $f \circ G$ is a path homotopy between $f \circ (e_0 * i) = e_{x_0} * f$ and $f \circ i = f$. The other identity uses $i * e_1$.

(3) $i * \bar{i}$ and e_0 are paths in I from 0 to 0, hence path homotopic by some H; then $f \circ H$ shows $f * \bar{f} \simeq_p e_{x_0}$. Similarly $\bar{f} * f \simeq_p e_{x_1}$.

(1) For $0 < a < b < 1$ let $k_{a,b}$ be the path that on $[0,a]$, $[a,b]$, $[b,1]$ is the positive linear map of that interval onto I followed by f, g, h respectively. Then $(f * g) * h = k_{1/2,3/4}$ and $f * (g * h) = k_{1/4,1/2}$. For any a, b, c, d, let $\gamma : I \to I$ be the piecewise linear map sending $0, a, b, 1$ to $0, c, d, 1$; then $k_{c,d} \circ \gamma = k_{a,b}$. Since γ and i are paths in I from 0 to 1, $\gamma \simeq_p i$, and composing with $k_{c,d}$ gives $k_{a,b} \simeq_p k_{c,d}$. ∎

Theorem 9.3 (51.3)

Let f be a path in X and $0 = a_0 < a_1 < \cdots < a_n = 1$. Let f_i be the positive linear map of I onto $[a_{i-1}, a_i]$ followed by f. Then $[f] = [f_1] * \cdots * [f_n]$.

Proof. By associativity the right side is well defined. Represent it by a specific bracketing of $f_1 * \cdots * f_n$; this path is defined on a subdivision $0 = b_0 < \cdots < b_n = 1$ and on $[b_{i-1}, b_i]$ is the linear map onto I followed by f_i. Hence it equals $f \circ \gamma$, where $\gamma : I \to I$ is the piecewise linear map sending each $[b_{i-1}, b_i]$ linearly onto $[a_{i-1}, a_i]$. Since $\gamma \simeq_p i$ in I, $f \circ \gamma \simeq_p f$. ■

9.2 The fundamental group

For $x_0 \in X$, a **loop** at x_0 is a path from x_0 to x_0. By Theorem 9.2 the path-homotopy classes of loops at x_0 form a group under $*$, the **fundamental group** $\pi_1(X, x_0)$. If X is a convex subset of $\mathbb{R}^n$, $\pi_1(X, x_0)$ is trivial (straight-line homotopy to the constant loop). For a path α from x_0 to x_1 define $\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$ by $\hat{\alpha}([f]) = [\bar{\alpha}] * [f] * [\alpha]$.

Theorem 9.4 (52.1)

$\hat{\alpha}$ is a group isomorphism.

Proof. $\hat{\alpha}([f]) * \hat{\alpha}([g]) = [\bar{\alpha}] * [f] * [\alpha] * [\bar{\alpha}] * [g] * [\alpha] = \hat{\alpha}([f] * [g])$. With $\beta = \bar{\alpha}$, $\hat{\beta}(\hat{\alpha}([h])) = [\alpha] * [\bar{\alpha}] * [h] * [\alpha] * [\bar{\alpha}] = [h]$, and similarly $\hat{\alpha} \circ \hat{\beta} = \mathrm{id}$. ■

Corollary 9.5 (52.2)

If X is path connected and $x_0, x_1 \in X$, then $\pi_1(X, x_0) \cong \pi_1(X, x_1)$.

If C is the path component of x_0, then $\pi_1(C, x_0) = \pi_1(X, x_0)$, since loops and homotopies based at x_0 lie in C. A path-connected space with trivial fundamental group is **simply connected**.

Lemma 9.6 (52.3)

In a simply connected space, any two paths with the same initial and final points are path homotopic.

Proof. If f, g run from x_0 to x_1, then $f * \bar{g}$ is a loop at x_0, so $[f] * [\bar{g}] = [e_{x_0}]$; multiplying on the right by $[g]$ gives $[f] = [g]$. ■

A continuous $h : (X, x_0) \to (Y, y_0)$ (that is, $h(x_0) = y_0$) induces $h_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$, $h_*([f]) = [h \circ f]$, well defined and a homomorphism by the observations before Theorem 9.2.

Theorem 9.7 (52.4)

If $h : (X, x_0) \to (Y, y_0)$ and $k : (Y, y_0) \to (Z, z_0)$ are continuous, then $(k \circ h)_* = k_* \circ h_*$. If i is the identity of (X, x_0), then i_* is the identity.

Proof. $(k \circ h)_*([f]) = [(k \circ h) \circ f] = [k \circ (h \circ f)] = k_*(h_*([f]))$, and $i_*([f]) = [f]$. ■

Corollary 9.8 (52.5)

If $h : (X, x_0) \to (Y, y_0)$ is a homeomorphism, h_* is an isomorphism.

Proof. With $k = h^{-1}$, $k_*h_* = (k \circ h)_* = \mathrm{id}$ and $h_*k_* = \mathrm{id}$. ■

9.3 Covering spaces

Let $p : E \to B$ be continuous and surjective. An open $U \subset B$ is **evenly covered** by p if $p^{-1}(U)$ is a union of disjoint open sets V_α (the **slices**), each mapped homeomorphically onto U by p. If every point of B has an evenly covered neighbourhood, p is a **covering map** and E a **covering space** of B. A covering map is a local homeomorphism and an open map, and each fibre $p^{-1}(b)$ is discrete (each of its points is the only one in its slice).

Theorem 9.9 (53.1)

$p : \mathbb{R} \to S^1$, $p(x) = (\cos 2\pi x, \sin 2\pi x)$, is a covering map.

Proof. Let U be the open arc of points with positive first coordinate. Then $p^{-1}(U) = \bigcup_n (n - \frac{1}{4}, n + \frac{1}{4})$. On $[n - \frac{1}{4}, n + \frac{1}{4}]$, p is injective (its second coordinate is strictly increasing), so it maps this compact interval homeomorphically onto the closed right half circle; restricting, $(n - \frac{1}{4}, n + \frac{1}{4})$ maps homeomorphically onto U. The same argument applies to the arcs with negative first coordinate, positive second coordinate and negative second coordinate, and these four arcs cover S^1. ■

Example 9.2

The restriction of p to $(0, \infty)$ is a surjective local homeomorphism but not a covering map: no neighbourhood of $b_0 = (1, 0)$ is evenly covered, because the component of $p^{-1}(U)$ containing a point near 0 is a half-open interval $(0, \epsilon)$ that is not mapped onto U.

Theorem 9.10 (53.2)

If $p : E \to B$ is a covering map, $B_0 \subset B$ and $E_0 = p^{-1}(B_0)$, then $p_0 = p|E_0 : E_0 \to B_0$ is a covering map.

Proof. For $b \in B_0$ let U be evenly covered with slices V_α. Then $U \cap B_0$ is a neighbourhood of b in B_0, and $p_0^{-1}(U \cap B_0)$ is the disjoint union of the sets $V_\alpha \cap E_0$, open in E_0, each mapped homeomorphically onto $U \cap B_0$. ■

Theorem 9.11 (53.3)

If $p : E \to B$ and $p' : E' \to B'$ are covering maps, so is $p \times p' : E \times E' \to B \times B'$.

Proof. If U, U' are evenly covered neighbourhoods of b, b' with slices V_α, V'_β, then $(p \times p')^{-1}(U \times U')$ is the disjoint union of the open sets $V_\alpha \times V'_\beta$, each mapped homeomorphically onto $U \times U'$. ■

Example 9.3

$p \times p : \mathbb{R} \times \mathbb{R} \to S^1 \times S^1$ is a covering of the torus $T = S^1 \times S^1$. Restricting to $b_0 \times S^1 \cup S^1 \times b_0$ (Theorem 9.10) gives a covering of the figure eight $b_0 \times S^1 \cup S^1 \times b_0$ by the grid of horizontal and vertical lines through integer points. Since $\mathbb{R}_+ \times S^1 \cong \mathbb{R}^2 - 0$ by polar coordinates, $\mathrm{id} \times p : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}^2 - 0$ is a covering of the punctured plane.

9.4 The fundamental group of the circle

If $p : E \to B$ is a map, a **lifting** of $f : X \to B$ is a map $\tilde{f} : X \to E$ with $p \circ \tilde{f} = f$.

Lemma 9.12 (54.1)

Let $p : E \to B$ be a covering map with $p(e_0) = b_0$. Any path $f : I \to B$ beginning at b_0 has a unique lifting $\tilde{f}$ beginning at e_0.

Proof. Cover B by evenly covered open sets; by the Lebesgue number lemma there is a subdivision $0 = s_0 < \cdots < s_n = 1$ such that each $f([s_i, s_{i+1}])$ lies in an evenly covered U.

Existence: set $\tilde{f}(0) = e_0$; if $\tilde{f}$ is defined on $[0, s_i]$, let V_0 be the slice of $p^{-1}(U)$ containing $\tilde{f}(s_i)$ and define $\tilde{f}(s) = (p|V_0)^{-1}(f(s))$ for $s \in [s_i, s_{i+1}]$; the pasting lemma gives continuity.

Uniqueness: if $\tilde{f}'$ is another lifting and $\tilde{f}' = \tilde{f}$ on $[0, s_i]$, then $\tilde{f}'([s_i, s_{i+1}])$ is connected and lies in the disjoint union of the open slices, hence in one slice, which must be V_0 since it contains $\tilde{f}'(s_i) = \tilde{f}(s_i)$. Then $\tilde{f}'(s) = (p|V_0)^{-1}(f(s)) = \tilde{f}(s)$ there. ∎

Lemma 9.13 (54.2)

Let $p : E \to B$ be a covering map with $p(e_0) = b_0$, and $F : I \times I \to B$ continuous with $F(0,0) = b_0$. There is a unique lifting $\tilde{F} : I \times I \to E$ with $\tilde{F}(0,0) = e_0$. If F is a path homotopy, so is $\tilde{F}$.

Proof. Existence: choose subdivisions $s_0 < \cdots < s_m$ and $t_0 < \cdots < t_n$ of I such that each rectangle $I_i \times J_j = [s_{i-1}, s_i] \times [t_{j-1}, t_j]$ is mapped by F into an evenly covered open set (Lebesgue number lemma). Define $\tilde{F}$ on $A = 0 \times I \cup I \times 0$ by lifting the two edges as paths from $(0,0)$ (Lemma 9.12). Then extend over the rectangles in the order $I_1 \times J_1, \ldots, I_m \times J_1, I_1 \times J_2, \ldots$. At each stage let A' be A together with the rectangles already treated, R the next rectangle and $C = R \cap A'$, the union of the left and bottom edges of R, which is connected. With $F(R) \subset U$ evenly covered, $\tilde{F}(C)$ is connected, so lies in one slice V_0; define $\tilde{F} = (p|V_0)^{-1} \circ F$ on R. This agrees with the earlier definition on C and is continuous on $A' \cup R$ by the pasting lemma. Uniqueness: any lifting is determined on A by Lemma 9.12, and on each successive R its image is connected, contains $\tilde{F}(C) \subset V_0$, hence lies in V_0, so it equals $(p|V_0)^{-1} \circ F$. If F is a path homotopy, $F(0 \times I) = b_0$, so $\tilde{F}(0 \times I)$ is a connected subset of the discrete fibre $p^{-1}(b_0)$, hence a point; likewise for $1 \times I$. ∎

Theorem 9.14 (54.3)

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$, and f, g paths in B from b_0 to b_1 with liftings $\tilde{f}, \tilde{g}$ beginning at e_0. If $f \simeq_p g$, then $\tilde{f}$ and $\tilde{g}$ end at the same point and $\tilde{f} \simeq_p \tilde{g}$.

Proof. Let $F : f \simeq_p g$ and $\tilde{F}$ its lifting with $\tilde{F}(0,0) = e_0$, a path homotopy (Lemma 9.13). $\tilde{F}|I \times 0$ lifts f and begins at e_0, so equals $\tilde{f}$; $\tilde{F}|I \times 1$ lifts g and begins at $\tilde{F}(0,1) = e_0$, so equals $\tilde{g}$. Since $\tilde{F}(1 \times I)$ is a point, $\tilde{f}(1) = \tilde{g}(1)$. ∎

The **lifting correspondence** $\phi : \pi_1(B, b_0) \to p^{-1}(b_0)$ is $\phi([f]) = \tilde{f}(1)$, where $\tilde{f}$ lifts f from e_0; it is well defined by Theorem 9.14.

Theorem 9.15 (54.4)

If E is path connected, ϕ is surjective; if E is simply connected, ϕ is bijective.

Proof. For $e_1 \in p^{-1}(b_0)$ take a path γ in E from e_0 to e_1; γ lifts the loop $f = p \circ \gamma$, so $\phi([f]) = e_1$. If $\phi([f]) = \phi([g])$ and E is simply connected, then $\tilde{f} \simeq_p \tilde{g}$ by Lemma 9.6, say by $\tilde{F}$, and $p \circ \tilde{F}$ is a path homotopy from f to g. ■

Theorem 9.16 (54.5)

$\pi_1(S^1, b_0)$ is isomorphic to the additive group $\mathbb{Z}$.

Proof. Use $p : \mathbb{R} \to S^1$ with $e_0 = 0$; $p^{-1}(b_0) = \mathbb{Z}$ and $\mathbb{R}$ is simply connected, so ϕ is a bijection onto $\mathbb{Z}$. If $\tilde{f}, \tilde{g}$ lift f, g from 0 with $\tilde{f}(1) = n$, $\tilde{g}(1) = m$, then $s \mapsto n + \tilde{g}(s)$ lifts g from n (as $p(n+x) = p(x)$), so $\tilde{f} * (n + \tilde{g})$ lifts $f * g$ from 0 and ends at $n + m$. Thus $\phi([f] * [g]) = \phi([f]) + \phi([g])$. ■

In particular the loop $p_0(s) = p(s)$ generates $\pi_1(S^1, b_0)$, and the class of a loop is determined by the end point n of its lifting, the number of times it winds around.

Theorem 9.17 (54.6)

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$.
(a) $p_* : \pi_1(E, e_0) \to \pi_1(B, b_0)$ is injective.
(b) Let $H = p_*(\pi_1(E, e_0))$. The lifting correspondence induces an injective map $\Phi : \pi_1(B, b_0)/H \to p^{-1}(b_0)$ of the right cosets of H, bijective if E is path connected.
(c) A loop f at b_0 has $[f] \in H$ iff f lifts to a loop at e_0.

Proof. (a) Let $\tilde{h}$ be a loop at e_0 with $p \circ \tilde{h} \simeq_p e_{b_0}$ by F. The lifting $\tilde{F}$ from e_0 is a path homotopy between the liftings of $p \circ \tilde{h}$ and e_{b_0} from e_0, namely $\tilde{h}$ and the constant loop.

(c) If $[f] = [p \circ \tilde{h}]$ with $\tilde{h}$ a loop at e_0, then by Theorem 9.14 $\tilde{f}$ ends where $\tilde{h}$ does, at e_0. Conversely, if $\tilde{f}$ is a loop, $[f] = p_*[\tilde{f}]$.

(b) Define $\Phi(H * [f]) = \phi([f])$. If $[h] \in H$, $h = p \circ \tilde{h}$ with $\tilde{h}$ a loop at e_0, then $\tilde{h} * \tilde{f}$ lifts $h * f$ and ends at $\tilde{f}(1)$; so Φ is well defined. If $\phi([f]) = \phi([g])$, then $\tilde{f} * \bar{\tilde{g}}$ is a loop at e_0 lifting $f * \bar{g}$, so $[f] * [g]^{-1} \in H$ and $H * [f] = H * [g]$. Surjectivity for path-connected E is Theorem 9.15. ■

9.5 Retractions and fixed points

$A \subset X$ is a **retract** of X if there is a continuous $r : X \to A$ with $r|A = \mathrm{id}_A$ (a **retraction**).

Lemma 9.18 (55.1)

If A is a retract of X, the homomorphism j_* induced by inclusion $j : A \to X$ is injective.

Proof. $r \circ j = \mathrm{id}_A$, so $r_* \circ j_*$ is the identity. ■

Theorem 9.19 (55.2, no-retraction theorem)

There is no retraction of B^2 onto S^1.

Proof. $j_* : \pi_1(S^1) \cong \mathbb{Z} \to \pi_1(B^2) = 0$ is not injective. ■

Lemma 9.20 (55.3)

For continuous $h : S^1 \to X$ the following are equivalent:
(1) h is nulhomotopic;
(2) h extends to a continuous $k : B^2 \to X$;
(3) h_* is the trivial homomorphism.

Proof. (1)⇒(2): let $H : S^1 \times I \to X$ be a homotopy from h to a constant. The map $\pi : S^1 \times I \to B^2$, $\pi(x,t) = (1-t)x$, is continuous, closed and surjective, hence a quotient map; it collapses $S^1 \times 1$ to 0 and is otherwise injective. H is constant on $S^1 \times 1$, so it induces a continuous k with $k \circ \pi = H$, and k extends h. (2)⇒(3): $h = k \circ j$ with $j : S^1 \to B^2$, and j_* is trivial. (3)⇒(1): let $p_0(s) = p(s)$ be the standard loop and $b_0 = p_0(0)$. Then $h \circ p_0$ is path homotopic to a constant by some $F : I \times I \to X$, with $F(0,t) = F(1,t) = h(b_0)$ and $F(s,1) = h(b_0)$. The map $q : I \times I \to B^2$, $q(s,t) = (1-t)p_0(s)$, is a quotient map that identifies $(0,t)$ with $(1,t)$ and collapses $I \times 1$; F respects these identifications, so it induces $k : B^2 \to X$ with $k \circ q = F$, and k extends h. Then $H(x,t) = k((1-t)x)$ is a homotopy from h to the constant $k(0)$. ■

Corollary 9.21 (55.4)

The inclusion $j : S^1 \to \mathbb{R}^2 - 0$ and the identity $i : S^1 \to S^1$ are not nulhomotopic.

Proof. $r(x) = x/\|x\|$ retracts $\mathbb{R}^2 - 0$ onto S^1, so j_* is injective (Lemma 9.18), hence nontrivial; i_* is the identity of $\mathbb{Z}$. Apply Lemma 9.20. ■

Theorem 9.22 (55.5)

Given a nonvanishing vector field on B^2 (a continuous $v : B^2 \to \mathbb{R}^2 - 0$), there is a point of S^1 where v points directly inward and one where it points directly outward.

Proof. Suppose $v(x)$ never points directly inward, i.e. $v(x) \neq ax$ for $x \in S^1$, $a < 0$. Then $w = v|S^1 : S^1 \to \mathbb{R}^2 - 0$ extends to B^2, so is nulhomotopic. $F(x,t) = tx + (1-t)w(x)$ never vanishes on $S^1 \times I$: for $t = 0, 1$ clearly, and for $0 < t < 1$, $F = 0$ would mean $w(x) = -\frac{t}{1-t}x$. So $j \simeq w$ in $\mathbb{R}^2 - 0$ and j is nulhomotopic, contradicting Corollary 9.21. For outward, apply this to $-v$. ■

Theorem 9.23 (55.6, Brouwer fixed-point theorem for the disc)

Every continuous $f : B^2 \to B^2$ has a fixed point.

Proof. Otherwise $v(x) = f(x) - x$ is a nonvanishing vector field on B^2. At $x \in S^1$ it cannot point directly outward, since $f(x) - x = ax$ with $a > 0$ gives $f(x) = (1+a)x \notin B^2$. This contradicts Theorem 9.22. ■

Corollary 9.24 (55.7)

A 3×3 matrix A of positive real numbers has a positive real eigenvalue.

Proof. Let $B = \{x \in S^2 : x_1, x_2, x_3 \geq 0\}$; radial projection $x \mapsto x/(x_1+x_2+x_3)$ is a homeomorphism of B onto a closed triangle, which is homeomorphic to B^2. For $x \in B$, Ax has positive entries, so $x \mapsto Ax/\|Ax\|$ maps B into B. A fixed point gives $Ax = \|Ax\|x$. ■

Theorem 9.25 (55.8)

Let T be a closed triangle in $\mathbb{R}^2$. There is $\epsilon > 0$ such that for every open covering $\mathcal{A}$ of T by sets of diameter less than ϵ, some point of T belongs to at least three elements of $\mathcal{A}$.

Proof. Since no point lies on all three edges e_1, e_2, e_3, the continuous function $(a, b, c) \mapsto \max(|a - b|, |b - c|, |a - c|)$ on the compact $e_1 \times e_2 \times e_3$ is positive and has a minimum $\epsilon > 0$; so no set of diameter less than ϵ meets all three edges. Suppose $\mathcal{A}$ is such a covering in which no point lies in three elements; passing to a finite subcovering, take a dominated partition of unity $\{\phi_A\}$. For each A choose a vertex v_A such that A does not meet the edge opposite v_A. Define $k(x) = \sum_A \phi_A(x) v_A$. Each x lies in at most two elements, so $k(x)$ is a convex combination of at most two vertices and lies in ∂T. If x lies on an edge e and $\phi_A(x) > 0$, then A meets e, so e is not opposite v_A, i.e. $v_A \in e$; thus $k(e) \subset e$. Then $H(x, t) = (1 - t)x + tk(x)$ is a homotopy in ∂T from the identity of ∂T to $k|\partial T$ (it stays in an edge containing x). But $k|\partial T$ extends to T, so is nulhomotopic (Lemma 9.20, $T \cong B^2$, $\partial T \cong S^1$), and then the identity of $\partial T \cong S^1$ would be nulhomotopic, contradicting Corollary 9.21. ■

So the triangle has topological dimension at least 2.

9.6 The fundamental theorem of algebra

Theorem 9.26 (56.1, fundamental theorem of algebra)

A polynomial equation $x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$ of degree $n > 0$ with real or complex coefficients has a complex root.

Proof.

Step 1. $f : S^1 \to S^1$, $f(z) = z^n$, induces an injective homomorphism: $f \circ p_0(s) = p(ns)$ lifts to $s \mapsto ns$, so f_* is multiplication by n on $\mathbb{Z}$.

Step 2. $g : S^1 \to \mathbb{R}^2 - 0$, $g(z) = z^n$, equals $j \circ f$, so $g_* = j_* f_*$ is injective and g is not nulhomotopic.

Step 3. if $\sum |a_i| < 1$, there is a root in B^2. Otherwise $k(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$ maps B^2 into $\mathbb{R}^2 - 0$, so $h = k|S^1$ is nulhomotopic. But $F(z, t) = z^n + t(a_{n-1}z^{n-1} + \cdots + a_0)$ never vanishes on $S^1 \times I$, since $|t(a_{n-1}z^{n-1} + \cdots + a_0)| \le \sum |a_i| < 1 = |z^n|$; so $g \simeq h$ in $\mathbb{R}^2 - 0$, a contradiction.

Step 4. in general, choose $c > 0$ with $\sum_i |a_{n-i}|/c^i < 1$ and substitute $x = cy$: the equation $y^n + (a_{n-1}/c)y^{n-1} + \cdots + a_0/c^n = 0$ has a root y_0, and cy_0 is a root of the original. ■

9.7 The Borsuk–Ulam theorem

$h : S^n \to S^m$ is **antipode-preserving** if $h(-x) = -h(x)$.

Theorem 9.27 (57.1)

A continuous antipode-preserving $h : S^1 \to S^1$ is not nulhomotopic.

Proof. Composing with a rotation (which commutes with $x \mapsto -x$) we may assume $h(b_0) = b_0$. Regard $S^1 \subset \mathbb{C}$ and let $q(z) = z^2$, a quotient map (continuous, closed, surjective) identifying z with $-z$. Since $q(h(-z)) = q(-h(z)) = q(h(z))$, there is a continuous $k : S^1 \to S^1$ with $k \circ q = q \circ h$. Now q_* is injective ($q \circ p_0$ lifts to $s \mapsto 2s$, so q_* is multiplication by 2). Let $f(s) = e^{\pi i s}$, a path from b_0 to $-b_0$; then $q \circ f = p_0$. The path $h \circ f$ runs from b_0 to $-b_0$, so its lifting $\tilde{g}$ from 0 ends at some

$m+\frac{1}{2}$; $q\circ h\circ f = p\circ 2\tilde{g}$, and $2\tilde{g}$ ends at the odd integer $2m+1$. So $k_*[p_0] = [k\circ q\circ f] = [q\circ h\circ f]$ corresponds to $2m+1\neq 0$, and k_* is injective. If h were nulhomotopic, h_* would be trivial, hence $k_*q_* = q_*h_*$ trivial, contradicting the injectivity of k_*q_*. ■

Theorem 9.28 (57.2)

There is no continuous antipode-preserving map $g : S^2 \to S^1$.

Proof. Its restriction h to the equator S^1 is antipode-preserving, hence not nulhomotopic. But h extends to the upper hemisphere, which is homeomorphic to B^2 by $(x,y,z)\mapsto(x,y)$; by Lemma 9.20 h is nulhomotopic. ■

Theorem 9.29 (57.3, Borsuk–Ulam theorem for S^2)

For every continuous $f : S^2 \to \mathbb{R}^2$ there is $x\in S^2$ with $f(x) = f(-x)$.

Proof. Otherwise $g(x) = (f(x)-f(-x))/\|f(x)-f(-x)\|$ is a continuous antipode-preserving map $S^2\to S^1$. ■

Theorem 9.30 (57.4, bisection theorem)

Given two bounded polygonal regions in $\mathbb{R}^2$, some line bisects each of them.

Proof. Put the regions A_1, A_2 in the plane $z=1$ of $\mathbb{R}^3$. For $u\in S^2$, let $f_i(u)$ be the area of the part of A_i on the side $x\cdot u > 0$ of the plane $P_u = \{x\cdot u = 0\}$. Then $f_i(-u) = \text{area}(A_i) - f_i(u)$, and f_i is continuous (we assume this elementary fact about areas). Borsuk–Ulam applied to (f_1, f_2) gives u with $f_i(u) = f_i(-u) = \frac{1}{2}\,\text{area}(A_i)$. Here $u\neq\pm(0,0,1)$, since then $f_i(u)$ is 0 or the full area; so P_u meets the plane $z=1$ in a line, which bisects both regions. ■

9.8 Deformation retracts and homotopy type

Lemma 9.31 (58.1)

If $h, k : (X,x_0)\to(Y,y_0)$ are homotopic by a homotopy keeping x_0 at y_0, then $h_* = k_*$.

Proof. For a loop f at x_0, $G(s,t) = H(f(s),t)$ is a path homotopy from $h\circ f$ to $k\circ f$. ■

Theorem 9.32 (58.2)

The inclusion $j : S^n \to \mathbb{R}^{n+1} - 0$ induces an isomorphism of fundamental groups.

Proof. $r(x) = x/\|x\|$ satisfies $r\circ j = \text{id}$, and $H(x,t) = (1-t)x + tx/\|x\|$ is a homotopy in $\mathbb{R}^{n+1}-0$ from the identity to $j\circ r$ fixing S^n pointwise. By Lemma 9.31, $j_*r_* = \text{id}$, and $r_*j_* = \text{id}$. ■

$A\subset X$ is a **deformation retract** of X if the identity of X is homotopic to a map carrying X into A by a homotopy fixing each point of A: $H : X\times I\to X$ with $H(x,0) = x$, $H(x,1)\in A$, $H(a,t) = a$ for $a\in A$.

Theorem 9.33 (58.3)

If A is a deformation retract of X and $x_0 \in A$, then $j : (A, x_0) \to (X, x_0)$ induces an isomorphism.

Proof. As in Theorem 9.32, with $r(x) = H(x, 1)$. ∎

Example 9.4

The doubly punctured plane $\mathbb{R}^2 - \{p, q\}$ deformation retracts onto a figure eight, and also onto a theta space; the solid torus $B^2 \times S^1$ and $\mathbb{R}^3$ minus the z-axis deformation retract onto a circle, so their fundamental groups are $\mathbb{Z}$.

$f : X \to Y$ is a **homotopy equivalence** with **homotopy inverse** $g : Y \to X$ if $g \circ f \simeq \mathrm{id}_X$ and $f \circ g \simeq \mathrm{id}_Y$; then X, Y have the same **homotopy type**.

Lemma 9.34 (58.4)

Let $h, k : X \to Y$ be homotopic by H, with $h(x_0) = y_0$, $k(x_0) = y_1$, and let $\alpha(t) = H(x_0, t)$. Then $k_* = \hat{\alpha} \circ h_*$.

Proof. Let f be a loop at x_0 and $G(s, t) = H(f(s), t)$ on $I \times I$. Consider the paths $\beta_0(s) = (s, 0)$, $\beta_1(s) = (s, 1)$, $\gamma_0(t) = (0, t)$, $\gamma_1(t) = (1, t)$ in $I \times I$. Both $\beta_0 * \gamma_1$ and $\gamma_0 * \beta_1$ run from $(0, 0)$ to $(1, 1)$ in the convex set $I \times I$, hence are path homotopic, and so are their images under G: $(h \circ f) * \alpha \simeq_p \alpha * (k \circ f)$. Thus $[k \circ f] = [\bar{\alpha}] * [h \circ f] * [\alpha]$. ∎

Corollary 9.35 (58.5)

If $h \simeq k$ and h_* is injective, surjective or trivial, so is k_*.

Corollary 9.36 (58.6)

If h is nulhomotopic, h_* is trivial.

Proof. Both follow from Lemma 9.34, since $\hat{\alpha}$ is an isomorphism and a constant map induces the trivial homomorphism. ∎

Theorem 9.37 (58.7)

If $f : X \to Y$ is a homotopy equivalence with $f(x_0) = y_0$, then $f_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$ is an isomorphism.

Proof. Let g be a homotopy inverse, $x_1 = g(y_0)$, $y_1 = f(x_1)$. Since $g \circ f \simeq \mathrm{id}_X$, Lemma 9.34 shows that $g_* \circ (f_{x_0})_* : \pi_1(X, x_0) \to \pi_1(X, x_1)$ is an isomorphism, so $(g_{y_0})_*$ is surjective. Since $f \circ g \simeq \mathrm{id}_Y$, $(f_{x_1})_* \circ (g_{y_0})_* : \pi_1(Y, y_0) \to \pi_1(Y, y_1)$ is an isomorphism, so $(g_{y_0})_*$ is injective. Hence $(g_{y_0})_*$ is an isomorphism, and so is $(f_{x_0})_*$. ∎

9.9 The fundamental group of S^n

Theorem 9.38 (59.1)

Let $X = U \cup V$ with U, V open, $U \cap V$ path connected and $x_0 \in U \cap V$. Then the images of $i_* : \pi_1(U, x_0) \to \pi_1(X, x_0)$ and $j_* : \pi_1(V, x_0) \to \pi_1(X, x_0)$ generate $\pi_1(X, x_0)$.

Proof. Let f be a loop at x_0. By the Lebesgue number lemma choose $0 = a_0 < \cdots < a_n = 1$ with each $f([a_{i-1}, a_i])$ in U or in V. If some $f(a_i)$ $(0 < i < n)$ is not in $U \cap V$, say it lies in $U - V$, then both adjacent intervals are mapped into U, and a_i can be deleted; repeating, we may assume $f(a_i) \in U \cap V$ for all i. Choose paths α_i in $U \cap V$ from x_0 to $f(a_i)$ (α_0, α_n constant). With f_i as in Theorem 9.3, $[f] = [f_1] * \cdots * [f_n] = [g_1] * \cdots * [g_n]$ where $g_i = \alpha_{i-1} * f_i * \bar{\alpha}_i$ is a loop at x_0 lying entirely in U or in V. ■

Corollary 9.39 (59.2)

If $X = U \cup V$ with U, V open and simply connected and $U \cap V$ nonempty and path connected, then X is simply connected.

Theorem 9.40 (59.3)

For $n \geq 2$, S^n is simply connected.

Proof. Let $p = (0, \ldots, 0, 1)$. Stereographic projection $f(x) = (x_1, \ldots, x_n)/(1 - x_{n+1})$ is a homeomorphism $S^n - p \to \mathbb{R}^n$, with inverse $y \mapsto (t(y)y, 1 - t(y))$, $t(y) = 2/(1 + \|y\|^2)$; similarly $S^n - (-p) \cong \mathbb{R}^n$. Both are simply connected, and their intersection is homeomorphic to $\mathbb{R}^n - 0$, which is path connected for $n \geq 2$. Apply Corollary 9.39. ■

9.10 Fundamental groups of some surfaces

A **surface** is a Hausdorff space with a countable basis in which each point has a neighbourhood homeomorphic to an open subset of $\mathbb{R}^2$.

Theorem 9.41 (60.1)

$\pi_1(X \times Y, x_0 \times y_0) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$.

Proof. With projections p, q, let $\Phi([f]) = (p_*[f], q_*[f])$, a homomorphism. Given loops g in X and h in Y, $f(s) = g(s) \times h(s)$ maps to $([g], [h])$. If $p \circ f \simeq_p e$ by G and $q \circ f \simeq_p e$ by H, then $G \times H$ is a path homotopy from f to the constant loop. ■

Corollary 9.42 (60.2)

The fundamental group of the torus $S^1 \times S^1$ is $\mathbb{Z} \times \mathbb{Z}$.

The **projective plane** P^2 is the quotient of S^2 obtained by identifying each x with $-x$.

Theorem 9.43 (60.3)

P^2 is a compact surface, and the quotient map $p : S^2 \to P^2$ is a covering map.

Proof. p is open: for U open, $p^{-1}(p(U)) = U \cup a(U)$ with $a(x) = -x$, which is open. For $x \in S^2$ let U be the ϵ-neighbourhood of x with $\epsilon < 1$; it contains no antipodal pair, so $p|U$ is a continuous open bijection onto $p(U)$, a homeomorphism, and similarly for $a(U)$, which is disjoint from U. So $p(U)$ is evenly covered with slices $U, a(U)$. Hausdorff: for $p(x) \neq p(y)$, the four points $\pm x, \pm y$ are distinct; take ϵ less than half their minimum distance, so the ϵ-neighbourhoods U of x and W of y satisfy $(U \cup a(U)) \cap (W \cup a(W)) = \emptyset$, whence $p(U) \cap p(W) = \emptyset$. A countable basis of S^2 maps to one of P^2 since p is open. Locally, P^2 is homeomorphic to open subsets of S^2, hence of $\mathbb{R}^2$; and it is compact as a continuous image of S^2. ■

Corollary 9.44 (60.4)

$\pi_1(P^2, y)$ has order 2.

Proof. S^2 is simply connected, so the lifting correspondence is a bijection onto the two-point fibre (Theorem 9.15). ■

Similarly P^n $(n \geq 2)$ has fundamental group of order 2.

Lemma 9.45 (60.5)

The fundamental group of the figure eight is not abelian.

Proof. Let $X = A \cup B$ be two circles meeting at x_0. Let E be the union of the coordinate axes in $\mathbb{R}^2$ together with, for each nonzero integer n, a circle tangent to the x-axis at $(n, 0)$ and a circle tangent to the y-axis at $(0, n)$. Define $p : E \to X$ by wrapping the x-axis around A (each $[n, n+1]$ once, integers to x_0), the y-axis around B, each circle tangent to the x-axis homeomorphically onto B and each circle tangent to the y-axis onto A (tangency points to x_0). Points of $X - x_0$ have small arc neighbourhoods whose preimages are disjoint arcs mapped homeomorphically, and x_0 has a small "cross" neighbourhood whose preimage is a disjoint union of crosses about the integer lattice points on the axes; so p is a covering map. Let g and h be loops going once around A and B. The lifting of $g * h$ from the origin runs along the x-axis to $(1, 0)$ and then around the circle tangent there, ending at $(1, 0)$; the lifting of $h * g$ ends at $(0, 1)$. By Theorem 9.14, $g * h \not\simeq_p h * g$. ■

Theorem 9.46 (60.6)

The fundamental group of the double torus is not abelian.

Proof. The double torus X contains a figure eight A, consisting of one circle around each handle meeting at a point, and A is a retract of X: each handle can be squashed continuously onto its circle, compatibly along the curve where the two halves are joined. By Lemma 9.18, $\pi_1(X)$ contains a copy of the nonabelian group $\pi_1(A)$. ■

Corollary 9.47 (60.7)

The 2-sphere, torus, projective plane and double torus are topologically distinct.

Proof. Their fundamental groups are trivial, $\mathbb{Z} \times \mathbb{Z}$ (infinite abelian), of order 2, and nonabelian. ■